PRODUCT INVENTORY MANAGEMENT SYSTEM

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1. Project Overview:

The Product Inventory Management System is a backend web application developed using Node.js, Express.js, and MongoDB.

It provides a secure solution for managing product data through RESTful APIs with JWT authentication.

Key Objectives

• Secure user authentication using JWT

• CRUD operations for product inventory

• User-specific product management

• Scalable backend using Node.js and MongoDB

Target Users

• Small business owners

• Freelancers and sellers

• Students learning backend development

• Developers practicing API development

2. Project Instructions:

• Create backend project folder

• Configure Express.js server

• Setup MongoDB database

• Implement authentication APIs

• Implement product CRUD APIs

• Test APIs using Thunder Client

[Insert Project Thumbnail Image Here]

3. Pre-Requisites:

Node.js – JavaScript runtime for backend

MongoDB – Database storage

npm – Package manager

VS Code – Development environment

Knowledge Requirements

• JavaScript ES6

• Node.js and Express.js

• MongoDB and Mongoose

• JWT Authentication

4. Project Architecture:

[Insert Architecture Diagram Here]

Technical Architecture

Controllers – Handle authentication and product logic

Models – Define database schema

Routes – API endpoints

Middleware – JWT verification

5. Database Architecture:

Users Collection: name, email, password, timestamps

Products Collection: name, price, quantity, description, userId

[Insert ER Diagram Here]

6. Core Features:

1. User Authentication

2. Product CRUD operations

3. Private user product access

4. Product search

7. Project Setup:

Initialize project using: npm init -y

Install packages:

express

mongoose

cors

dotenv

jsonwebtoken

bcryptjs

[Insert Folder Structure Image Here]

8. Project Execution:

1. Open terminal in server folder

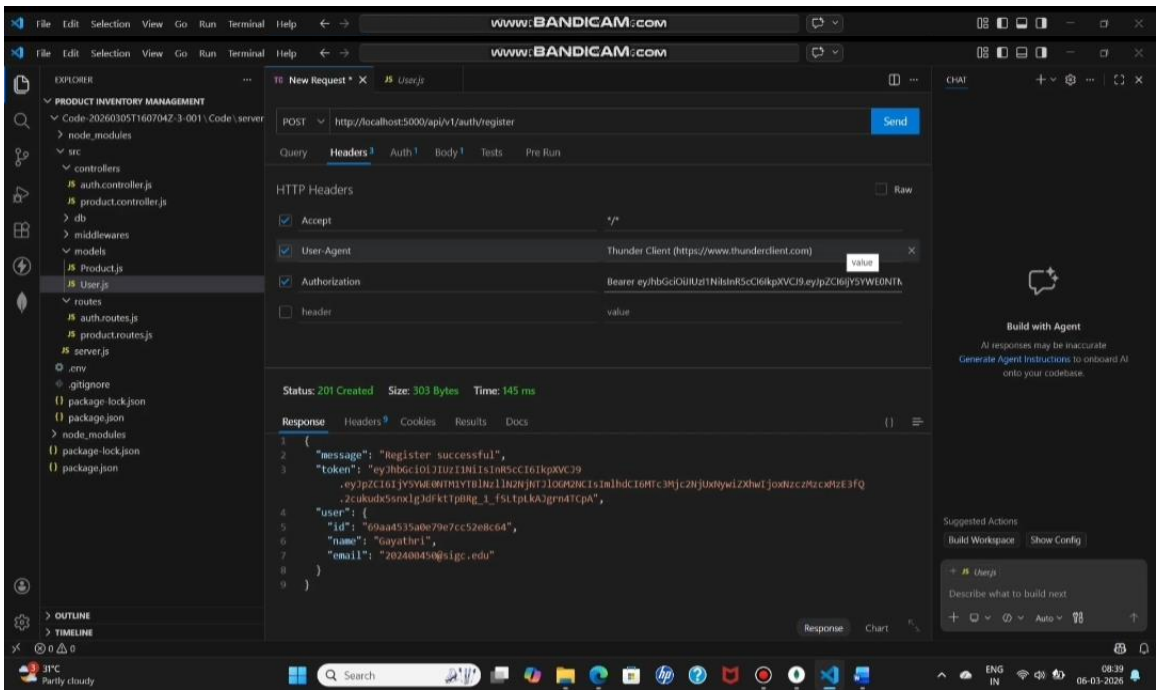
2. Run npm start

3. Server connects to database9. API Testing:

Use Thunder Client to test APIs

DEMO VIDEO LINK :
https://drive.google.com/file/d/1Z8Y0KMnA8ZK7zOn_GM5g9sN3pfsYdwx9/view?usp=drivesdk

[Insert Register API Screenshot Here]



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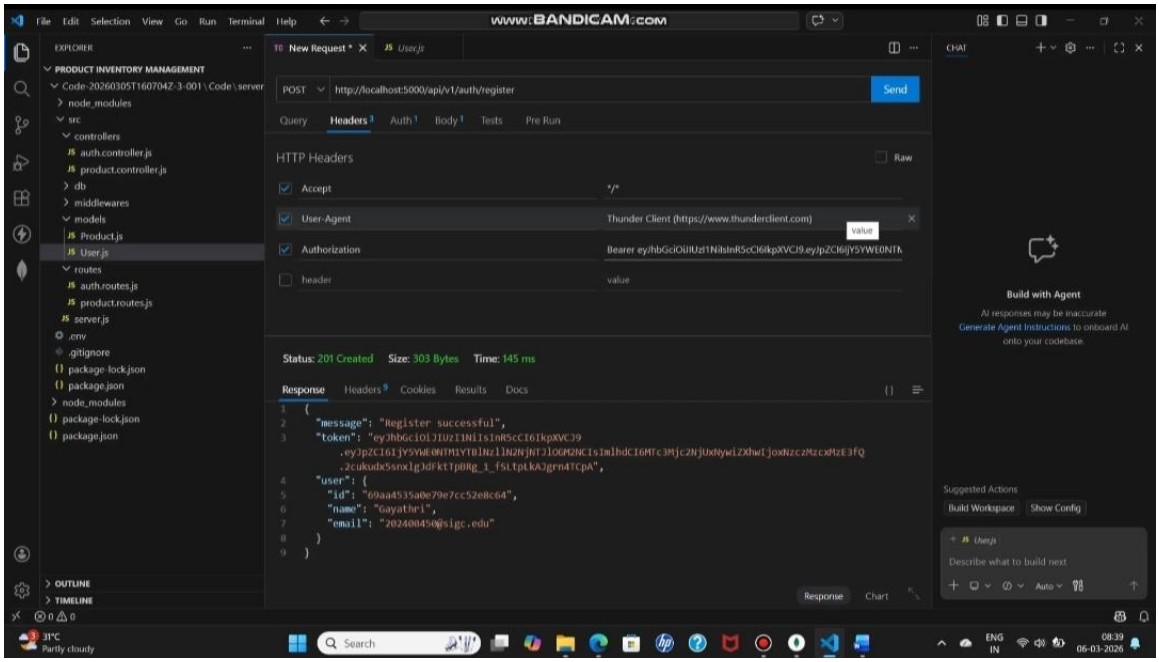
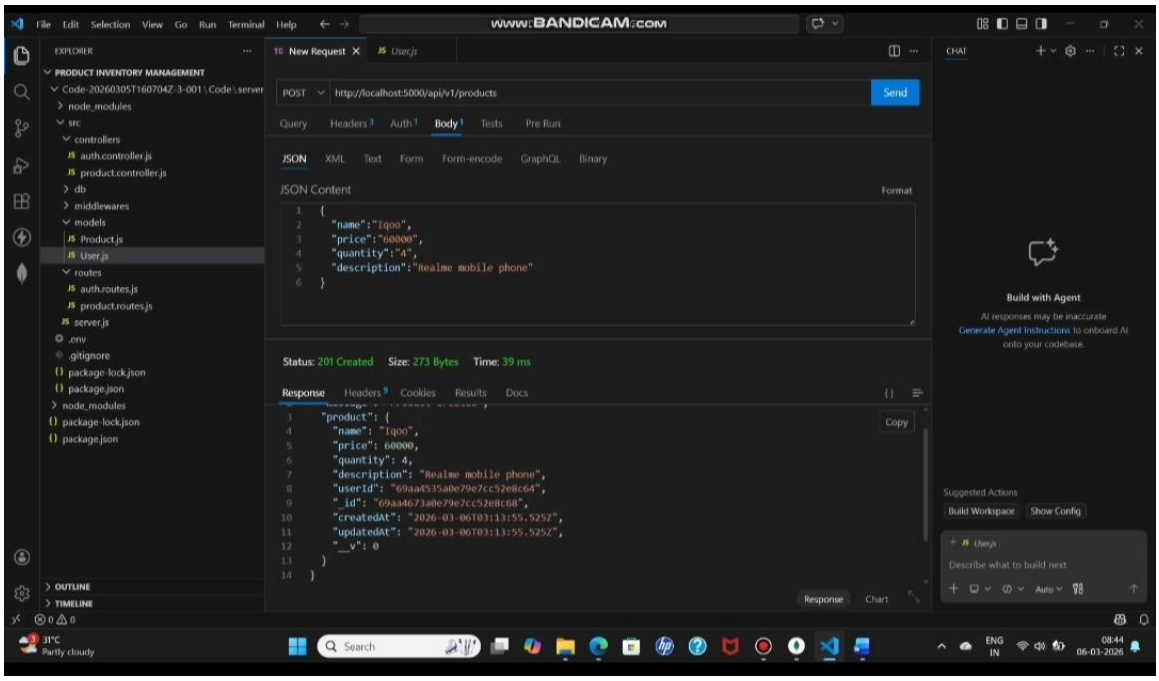


Insert Login API Screenshot Here]

[Insert Get Products Screenshot Here]